

instruction fed to the policy, and the winning index selects ASC’s target and threshold.

5 Experiments

The main census is the complete LIBERO-10 partition of LIBERO-Plus: 2,519 specifications, one trial each, enumerated in official identifier order with initial-state index 0, of which 393 are RI specifications (dose labels and rounding conventions: Appendix A). The section proceeds from the state operator’s main effect and its decomposition (Sections 5.1–5.2) through composed perturbations and transfer to independently trained policies (Sections 5.3–5.4) to the language operator and the deployed routed configuration (Sections 5.5–5.6).

Two evidence tiers apply throughout. The full-partition census tables and the banked three-training-seed cohort are reported as descriptive single-execution-block evaluations; intervals are attached only to paired comparisons within one execution block, and to preregistered multi-seed aggregates whose first-level resampling unit is the training seed. Current-implementation comparisons use one checkpoint and stateless common random numbers (CRN) for each specification, trial, and replan within an execution block, and report paired differences with base-task-clustered 95% bootstrap intervals; results from different files, evaluation seeds, controller generations, or execution blocks are not pooled. Paired replication of identically configured episodes additionally measures the protocol’s own nondeterminism floor, 188 of 1,791 replicated no-op episodes diverging in action stream and 24 flipping outcome (Appendix D), and per-factor differences of comparable size are read against this floor. That figure is conditional on the operator not having acted: a null replicate that alters only the numerical route to an identical servo target diverges on every episode where ASC fired and flips 26 of 99 (Appendix G), so it does not transfer to small samples dominated by treated episodes.

5.1 State Canonicalization: Main Effect and Target Attribution

ASC removes most of the RI deficit on this census. Table 1 reports four configurations under stateless common random numbers with one training seed; the first three share one execution block, and the routed row is a separate paired rerun (Section 5.6). With the task index provided, ASC raises overall SR and closes most of the RI deficit, from 40.20 to 76.08. The ladder also identifies the active ingredient: the frame-correct task-agnostic home (Section 4.1) recovers most of that deficit on its own, so servoing the arm back to a start pose expressed in the correct frame is the dominant effect. The task-agnostic and task-index configurations differ in target identity and in deadband, so they intervene on nested episode sets (374 of the 728 task-index triggers), and the remaining step covers both; a preregistered rerun at matched intervention frequency separates them next.

A single pooled start pose carries the census-wide effect. Within the same paired campaign as the comparison of Section 5.6, we rerun the task-index configuration alongside a matched-gate control that keeps the task-index gate and deadbands but replaces every per-task recovery target with the single pooled task-agnostic target. The two

Table 1: Main table: four configurations on the complete LIBERO-10 partition of LIBERO-Plus (2,519 specifications, one trial each; one training seed; evidence tiers: Section 5). The first three share one execution block. SR in percent by factor. [†]MiniLM-routed row: separate CRN-paired rerun (Section 5.6), not pooled with the first three rows. Rounding conventions: Appendix A.

Recovery	BG	Camera	Lang.	Light	Layout	Sensor	RI	ALL
No recovery	49.48	82.58	67.89	71.90	75.64	84.63	40.20	68.28
Global home	48.10	82.58	68.15	74.45	75.64	83.96	71.25	73.16
(task-agnostic)								
Task index	52.60	83.29	72.85	75.91	75.96	84.63	76.08	75.59
MiniLM routed [†]	50.17	82.82	90.34	75.18	76.28	80.40	73.79	76.74

In-run routing accuracy for the MiniLM-routed row is $368/383 = 96.08\%$ on the Language factor, which carries every misroute; the remaining 2,136 rows carry canonical instructions that route to themselves by construction, for $2,504/2,519 = 99.40\%$ over all rows.

runs make identical gate decisions on all 2,519 specifications (verified row by row against a CPU-precomputed gate certificate), so the contrast isolates target identity at fixed intervention frequency. The preregistered primary estimand, the paired difference over all 2,519 specifications, falls below the preregistered acceptance criterion for a real increment (Table 12): at matched intervention frequency, the single pooled start pose already delivers the census-wide effect, and most of the banked +2.42-point overall step from the task-agnostic to the task-index configuration is attributable to gate width. The RI slice, declared secondary with no acceptance criterion before the data were seen, moves +4.58 points with a one-sided lower bound of +0.98, but its exact McNemar p of 0.054 does not resolve at the conventional level (Table 12); it is directionally consistent with the matched-trigger diagnostic (trigger-set decomposition and divergence analysis: Appendix D).

5.2 The State Effect Reproduces Across Seeds and Implementations

The main effect is not a property of one training seed. On the fixed 250-specification subset with three trials per specification, a banked task-index-provided cohort that predates the current controller implementation spans three training seeds: ASC raises RI by between 28.7 and 36.1 points across the three seeds and overall success by four to five points, with no seed reversing either sign (Table 26 in Appendix H).

Nor is it a property of one controller implementation. Rerunning the seed-1 checkpoint with matched implementation and stateless common random numbers, task-index-provided ASC improves overall SR in block 0 by an amount close to the banked cohort’s, with a base-task-clustered 95% interval excluding zero (Table 27); it is not pooled with the banked seed-1 row.

The gate buys intervention economy. Gating task-index-provided ASC cuts controller use to about a fifth of the always-on setting, in episodes fired and in total controller steps alike (Table 28), while the always-on setting’s paired difference from the gated one has an interval covering zero (Table 27). Thus “active” denotes selective intervention; the success-rate effect comes from the recovery motion, not from the gate itself (Appendix E). Triggering outside RI is